



1) All of the following are examples of integral data types EXCEPT _____.

- ☐ A Double
- ☐ B Char
- ☐ C Short
- ☐ D Int

2) After the execution of the following code, what will be the value of numb if the input value is 5?

```
1. cin >> numb;  
2. if (numb > 0)  
3.   numb = numb + 10;  
4. else  
5.   if (numb > 5)  
6.     numb = numb + 15;
```

- ☐ A 0
- ☐ B 5
- ☐ C 25
- ☒ D 15

3) What type of C++ statement(s) stores a value in a variable?

- ☒ A both input and assignments
- ☐ B assignments
- ☐ C output
- ☐ D input

4) Which of the following is NOT a reserved word in C++?

- ☐ A return
- ☐ B int
- ☐ C const
- ☒ D num

5) What is the output?

```
1. int n = 0;  
2. cout << ((n) ? n++ : ++n);
```

- ☐ A 0
- ☒ B 1
- ☐ C None of these
- ☐ D 2



6) What is the output?

```
1. int x = 5;
2. cout << (x++) + x << endl;
```

- ☐ A 11
- ☐ B 10
- ☐ C None of these
- ☐ D 12

7) What is the output of the following code?

```
1. if ( 6 > 8)
2. {
3. cout << " *** " << endl ;
4. cout << "*****" << endl;
5. }
6. else if (9 == 4)
7. cout << "*****" << endl;
8. else
9. cout << " *" << endl;
```

- ☐ A *****
- ☐ B ***
- ☒ C *
- ☐ D **

8) Which of these is a valid reserved word in C++?

- ☐ A Char
- ☐ B Include
- ☐ C Double
- ☐ D Const
- ☒ E None of these

9) Suppose that x, y, z, and w are int variables. The expression $x(y+z)/w$ in C++ is written as

- ☒ A $x * (y + z) / w$
- ☐ B $x * y + z / w$
- ☐ C $x * y + x * z / w$
- ☐ D $x (y + z) / w$

10) The expression `static_cast<int>(6.9) + static_cast<int>(7.9)` evaluates to ____.

- ☐ A 14
- ☐ B 14.8
- ☒ C 13
- ☐ D 15



11) Suppose that ch1, ch2, and ch3 are variables of the type char and the input is:

A B
C

**What is the value of ch3 after the following statements execute?

```
1. cin.get(ch1);  
2. cin.get(ch2);  
   cin.get(ch3);
```

- ☒ A 'B'
- ☐ B 'C'
- ☐ C 'A'
- ☐ D '\n'

12) Suppose x and y are int variables. Consider the following statements:

```
1. if (x > 5)  
2. y = 1;  
3. else if (x < 5)  
4. {  
5.   if (x < 3)  
6.     y = 2;  
7.   else  
8.     y = 3;  
9. }  
10. else  
11. y = 4;
```

**What is the value of y if x = 3?

- ☒ A 3
- ☐ B 1
- ☐ C 2
- ☐ D 4



13) The value of the expression $26 - 14 \% 3 + 1$ is ____.

- ☐ A 1
- ☒ B 25
- ☐ C 24
- ☐ D 0

14) Suppose that x is an int variable, ch is a char variable, and the input is: 276.

**Choose the values after the following statement executes: `cin >> ch >> x;`

- ☒ A `ch = '2', x = 76`
- ☐ B `ch = '2', x = 276`
- ☐ C `ch = ' ', x = 276`
- ☐ D Input Failure

```
1. char ch1, ch2;
2. cin >> ch1;
3. cin.get(ch2);
4. cout << ch1 << ch2;
```

15) If the input is: M N

**what is the output?

- ☐ A MN
- ☒ B M
- ☐ C M N
- ☐ D None of these
- ☐ E N

16) What is the output of the following C++ code?

```
1. x = 6;
2. if (x > 10)
3. cout << "One ";
4. cout << "Two ";
```

- ☐ A One
- ☒ B Two
- ☐ C One Two
- ☐ D Two One



17) What is the output?

```
1. int x = 1;
2. while (--x)
3. cout << x;
```

- ☐ A 0
- ☐ B 1
- ☒ C No output
- ☐ D Infinite loop

18) Which of the following operators has the highest precedence?

- ☐ A *
- ☒ B !
- ☐ C %
- ☐ D =

19) If the input is: 35 62&1.78

**what are the values of x,z,ch?

```
1. int x;
2. double z;
3. char ch;
4. cin >> x >> ch >> z;
```

- ☒ A x = 35, ch = '6', z = 2.0 (or 2)
- ☐ B x = 35, ch = 6, z = 2.0
- ☐ C Input Failure
- ☐ D x = 35, ch = 62, z = 78

20) What is the output?

```
1. int n = 0;
2. cout << ((n) ? n++ : ++n);
```

- ☒ A 1
- ☐ B 0
- ☐ C None of these
- ☐ D 2



21) After the execution of the following code, what is the value of sum?

```
1. int sum = 0;
2. int num = 10;
3. if (num > 0)
4.     sum = sum + 10;
5. else if (num > 5)
6.     sum = num + 15;
```

- ☐ A 20
- ☒ B 10
- ☐ C 25
- ☐ D 0

22) Assume you have three int variables: x = 2, y = 6, and z. Choose the value of z in the following expression:

```
z = (y / x > 0) ? x : y;
```

- ☒ A 2
- ☐ B 3
- ☐ C 6
- ☐ D 4

23) What is the output?

```
1. cout << static_cast<int>(7.5 + static_cast<double>(19 / 2));
```

- ☒ A 16
- ☐ B 16.5
- ☐ C None of these
- ☐ D 17
- ☐ E 17.5



24) Which of the following is a reserved word in C++?

- ☐ A Char
- ☐ B include
- ☒ C void
- ☐ D duple

25) Which of these is an **invalid identifier**?

- ☐ A None of these
- ☐ B pop_
- ☐ C _D3
- ☒ D 1b
- ☐ E _sum3

26) Which of the following is a legal identifier?

- ☐ A 1program
- ☒ B program_1
- ☐ C program 1
- ☐ D program!

27) What is the output?

```
1. cout<<"Welcome\rHi";
```

- ☐ A Syntax Error
- ☒ B Hilcome
- ☐ C Hi
Welcome
- ☐ D HiWelcome



28) What is the output?

```
1. int i = 4, j = 6, k = 7, n = 3;  
2. cout << i + j*k - k%n << endl;  
3. cout << i / k << endl;
```

- ☐ A 45
0.66667
- ☐ B None of these
- ☐ C 0
- ☒ D 45
0

29) What is the output?

```
1. int x = 5;  
2. if (x-- == 4)  
3. cout << "One" << endl;  
4. cout << x << endl;  
5. cout << "Two";
```

- ☐ A 5
Two
- ☐ B One
Two
- ☐ C One
5 Two
- ☒ D 4
Two
- ☐ E One
4 Two



30) The conditional operator `? :` takes ____ arguments.

- ☐ A 1
- ☐ B 2
- ☒ C 3
- ☐ D 4

31) What is the output?

```
1. if (15 % 5)
2.   cout << "Two";
3. else cout << "One";
```

- ☒ A One
- ☐ B None of these
- ☐ C Two
- ☐ D Two
One

32) Which of the following is the new line character?

- ☐ A '\r'
- ☐ B '/n'
- ☒ C '\n'
- ☐ D '/r'

33) Which of the following expressions correctly determines that x is greater than 10 and less than 20?

- ☐ A $10 < x < 20$
- ☐ B $(10 < x < 20)$
- ☒ C $10 < x \ \&\& \ x < 20$
- ☐ D $10 < x \ || \ x < 20$



34) What is the output?

```
1. if (1 == 4 >= 1)
2.   cout << "One";
3. else
4.   cout << "Two";
```

- ☐ A OneTwo
- ☒ B One
- ☐ C Two
- ☐ D None of these

35) Which of these will **cause a syntax error**?

- ☐ A float f = 1000;
- ☐ B char ch = ' ';
- ☐ C bool b = 111;
- ☒ D None of these

36) What is the output of the following code fragment?

```
1. int x = 10;
2. if (x > 15)
3.   x = 0;
4.   cout << x << endl;
5. else
6.   cout << x + 5;
```

- ☐ A 5
- ☐ B 10
- ☐ C 0
- ☒ D None of these



37) Suppose x and y are int variables. Consider the following statements:

```
1. if (x > 5)
2.   y = 1;
3. else if (x < 5)
4.   {
5.     if (x < 3)
6.       y = 2;
7.     else
8.       y = 3;
9.   }
10. else
11.   y = 4;
```

****What is the value of y if x = 6?**

- ☒ A 1
- ☐ B 3
- ☐ C 2
- ☐ D 4

38) What is the output?

```
1. int n = 1;
2. cout << ((n) ? n++ : ++n);
```

- ☐ A 3
- ☒ B 1
- ☐ C 2
- ☐ D None of these



39) Which of these is a valid use of c?

1. `const int c = 9;`

- ☐ A `c = c+1;`
- ☒ B `cout<< c*3;`
- ☐ C `c = 20;`
- ☐ D None of these
- ☐ E `cin>>c;`

40) Which of the following is a relational operator?

- ☒ A `==`
- ☐ B `&&`
- ☐ C `=`
- ☐ D `!`

41) Suppose that x and y are int variables, z is a double variable, and the input is:

28

32.6 12

**Choose the values of x, y, and z after the following statement executes:

```
cin >> x >> y >> z;
```

- ☐ A `x = 28, y = 12, z = 32.6`
- ☒ B `x = 28, y = 32, z = 0.6`
- ☐ C `x = 28, y = 12, z = 0.6`
- ☐ D `x = 28, y = 32, z = 12.0`



42) Suppose that alpha and beta are int variables. The statement `alpha = beta-- ;` is equivalent to the statement(s)

- ☐ A `alpha = beta - 1;`
`beta--;`
- ☐ B `alpha = 1 - beta;`
`beta--;`
- ☐ C None of these
- ☒ D `alpha = beta;`
`beta = beta - 1;`

43) Consider the following statement:

```
int y = !(12 < 5 || 3 <= 5 && 3 > x) ? 7 : 9;
```

**What is the value of y if x = 2?

- ☐ A 3
- ☐ B 2
- ☐ C 7
- ☒ D 9

44) Which of the following will cause a **semantic error**, if you are trying to compare x to 5?

- ☒ A `if (x = 5)`
- ☐ B `if (x >= 5)`
- ☐ C `if (x == 5)`
- ☐ D `if (x <= 5)`



45) What is the value of x after the following statements execute?

```
1. int x;
2. x = (5 <= 3 && 'A' < 'F') ? 3 : 4;
```

- ☒ A 2
- ☐ B 3
- ☐ C 5
- ☐ D 4

46) In a ____ control structure, the computer executes particular statements depending on some condition(s).

- ☐ A repetition
- ☐ B looping
- ☒ C selection
- ☐ D sequence

47) What is the output of the following C++ code?

```
1. int x = 55, y = 5;
2. switch (x % 7)
3. { case 0: case 1: y++;
4. case 2: case 3: y = y + 2;
5. case 4: break;
6. case 5: case 6: y = y - 3;}
7. cout << y << endl;
```

- ☒ A 2
- ☐ B 5
- ☐ C 8
- ☐ D None of these

48) Consider the following statement:

```
int y = !(12 < 5 || 3 <= 5 && 3 > x) ? 7 : 9;
```

**What is the value of y if x = 2?

- ☐ A 2
- ☐ B 3
- ☐ C 7
- ☒ D 9



49) A(n) ____ statement causes an immediate exit from the **switch** structure.

- ☒ A break
- ☐ B endl
- ☐ C default
- ☐ D exit

50) After the execution of the following code, what is the value of sum?

```
1. int sum = 0;
2. int num = 10;
3. if (num > 0)
4. sum = sum + 10;
5. else if (num > 5)
6. sum = num + 15;
```

- ☐ A 20
- ☒ B 10
- ☐ C 25
- ☐ D 0

51) Consider the following code.

```
1. const int LASTVAL = -99;
2. int entry;
3. cin >> entry;
4. while (entry != LASTVAL)
5. {
6. triple = entry * 3;
7. cout << triple << endl;
8. cin >> entry;
9. }
```

****This code is an example of a(n) ____ while loop.**

- ☐ A EOF-controlled
- ☐ B counter-controlled
- ☒ C sentinel-controlled
- ☐ D None of answers.



52) Assume all variables are properly declared. What is the output of the following C++ code?

```
1. num = 100;
2. while (num <= 150)
3. num = num + 5;
4. cout << num << endl;
```

- ☐ A 160
- ☐ B 150
- ☐ C 145
- ☒ D 155

53) What is the output?

```
1. int x = 5;
2. cout << (x++) + x << endl;
```

- | | |
|---------------------------------------|----------------------------|
| ☒ A 10 | ☐ B 12 |
| ☐ C None of these | ☐ D 11 |

54) What is the output of the following program segment?

```
1. int x = 14, y = 60;
2. while (((y - x) % 3) != 0)
3. {
4. cout << y << " ";
5. y = y - 5;
6. }
```

- | | |
|--|-------------------------------------|
| ☒ A 60 55 | ☐ B 46 27 |
| ☐ C 60 55 50 | ☐ D 60 55 50 45 |



55) Which of the following is true about a while loop?

- ☐ A It cannot contain if statement.
- ☒ B The body of the loop may not execute at all.
- ☐ C The body of the loop is executed at least once.
- ☐ D The logical expression controlling the loop is evaluated before the loop is entered and after the loop exits.

56) What is the output of the following C++ code?

```
1. count = 1;  
2. num = 25;  
3. while (count < 25)  
4. { num = num - 1;  
5. count++;}  
6. cout << count << " " << num << endl;
```

- ☒ A 25 1
- ☐ B 24 1
- ☐ C 24 0
- ☐ D 25 0

57) In ____ structures, the computer repeats particular statements a certain number of times depending on some condition(s).

- ☒ A looping
- ☐ B branching
- ☐ C sequence
- ☐ D selection

58) What is the output of the following C++ code?

```
1. num = 0;  
2. while (num < 5)  
3. { cout << num << " ";  
4. num = num + 1; }
```

- ☐ A 0 1 2 3 4 5
- ☒ B 0 1 2 3 4
- ☐ C 1 2 3 4 5
- ☐ D 1 2 3 4 0



59) Suppose sum and num are int variables, and the input is 18 25 61 6 -A What is the output of the following code?

```
1. sum = 0;  
2. cin >> num;  
3. while (num != -1)  
4. {  
5. sum = sum + num;  
6. cin >> num;  
7. }  
8. cout << sum << endl;
```

- ☒ A 110
- ☐ B 92
- ☐ C 109
- ☐ D 111

^_^ GOOD LUCK ^_^